

Lec 01 Introduction to Generative AI

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Transcript

[00:00] Hello everyone, welcome to the lecture series on fundamentals of generative AI and the large language models. This is Ashini tutor for this uh particular course. Uh and throughout this particular course we will be learning the concepts on autoencoders, variational autoencoders, generative adversarial networks, GAN diffusion models. Then once we complete this we will be having uh uh some introduction to sequence-to-sequence models where we'll be learning RNN LSTMs so that it will act as a foundation for LLMs going forward. Next after sequence-to-sequence models we will learn LLMs then we look into RAG then multimodal AI and ethics. So these are uh some of the concepts that we will be covering

[00:53] throughout this particular 12 weeks course along with the theoretical concept you will also be seeing the hands-on implementation. So before we actually get into the concepts of generative AI this week one we will be completely focusing on the fundamentals of deep learning which are required for generative AI. So these are the following contents for week one. First we will understand uh uh so far we have seen in the deep learning concepts we have seen something uh called as discriminative models. So we now we will be understanding the shift from discriminative learning

to generative learning. Then we will understand what exactly is the generative modeling. What are different types of generative

[01:39] modeling. Then we will look into the probabil probability distribution in generative models. It can be a gaussian, bernoulli or it can be a categorical. Then let us look into some of the applications of generative AI where all we can see generative AI models. Then uh before we actually start the contents of generative AI, we will see some of the we will look into some of the basics such as what are various loss functions that are there in uh deep learning. what are different uh activation functions and optimizers various optimizers that are required in neural network the and um most of the generative AI models which we are focusing in this particular course uh some of them are based on CNN.

[02:30] So how CNN works um these generative models architecture also follows a similar pattern. So because of this we will first uh look quickly a recap of how CNN works. Towards the end of this particular week we will look into the hands-on implementation since in this whole week the key concept here is of course we will be seeing some of the fundamentals of generative AI and then we will look into the basics of deep learning and we will quickly recap CNN.

[03:02] CNN is very much needed so that upcoming models if some of the concepts that are based on this CNN probably it will be helpful for the learners to connect to the concept. So because of this the hands-on will be based on CNN and then we will see two hands-on in this particular week. One is building CNN on real world data set and the second how do you build an ensemble model. So these are the contents which we are going to study in this particular uh week. So let us understand uh where are we coming from. So if you see uh so far we have looked into this is multi-layer perceptron. Uh this is the architecture of multi-layer perceptron uh model. So [snorts] if you see this particular model works very well uh if you have an

[03:51] applications such as a classification and a regression problem and more specifically if we are working on a tabulated data where you have your data in the form of a CSV file or an excel file so on that particular data set if you want to perform a classification or a regression problem regression task then a simple multi-layer perceptron model is sufficient to perform the task. Suppose we have an image uh data set where our data is an images. So whenever we have an images and if our task is to do a classification or if our task is to do a segmentation or an object detection then probably uh the most appropriate algorithm will be a CNN convolutional neural network mainly in order to understand how the images

[04:44] works. uh in order to understand the images and then perform tasks such as a classification then uh segmentation or an object detection and the third model that we have seen so far is um you know sequence to sequence models. So uh examples for sequence to sequence is RNN and LSTMs or it can be even GRU. So these are the

models which works very well if you have a time dependent data uh time series data uh and uh what is the main application of this we can able to predict what will be our next uh data. If we have a last 10 days uh uh data then we can predict what will be our tomorrow's uh task or what will be tomorrow's value. If we have last 10 days uh rainfall prediction then we can say what is the probability that

[05:37] tomorrow we get rain. So these are some of the applications that have that we have seen in our conventional deep learning uh uh concepts and uh these are for all of these applications for all of these uh models how is that the data that we have considered is it not we had a labeled data set. What do you mean by labeled a data set? Here we had an uh sample and we also had an associated label with this. We had an input and we had an associated label. So uh and here what is X? X is input and this input can be an image or it can be a text embeddings or it can be a simple tabulated features or time series. It can be any of these. And what is more specifically y here and y here uh is a label or a target uh class. So when we

[06:34] say label um it can be an numerical value if it is a regression problem or it can be some category if it is an class label if it is an classification problem. So what are these models? Whatever we have seen in uh this particular slide take multi-layer perceptron CNN or RNN LSTMs. All these are discriminative models. So what exactly is a discriminative models? Look at these two diagrams here. Discriminative models try to learn to discriminate between the classes. Now if you take it is a cat and here this is a dog. Now when we look into this particular data, we can clearly discriminate between these two classes. How are we discriminating here?

[07:23] I'm going to learn the features. Okay, if it is a dog, this is uh this is how the features are. This is how the features. So we are going to learn each of these features. And if it is a cat, look at uh the features here. And definitely the features of this class and this class are totally different. So discriminative models are those models which learn to discriminate between the classes how it learns the features. By learning the feature it is going to separate one output from another output and it is just not only discriminating um or learning the features. it is learning from all of the training data and for each of the training data there is also a class label associated with it. So because we have a features and we

[08:14] have a labels we can able to learn these features and for each of these features what is its associated class. So this is discriminative model mainly focusing on discriminating between the classes by learning the uh features. Uh and for each of these features we have an correct answer already present or already known to it. Uh now uh what is one drawback of discriminative models? It its main focus is only to learn the features and discriminate. So it cannot learn how the data was generated, how these features are obtained. So discriminative model never learns how the data was generated. So what is its main aim? Discriminative models mapping input to an output. So if this is a kind of an input that we have then this is

[09:09] the uh label. So this is what is the learning. So more specifically if we have to say what is uh uh given this input uh x what uh what is the most likely the label or the output y what is the probability that you predict uh y for a given input x. So this is what a discriminative model does. So one example if you have to take suppose X is an image let us take that image is a cat image dog image and if it is a multiclass we have an horse also so then what is that model will give you the output what is the probability that a given input X belongs to this particular class cat so the probability is 0.85 or it is 85%. What is the probability that the given input belongs to a dog class?

[10:02] It is 0.10. Similar way 0.05 horse. So here what exactly from this particular output we are understanding is it not the model is mainly aiming to learn which class the particular input belongs to. Right? So this is what is the discriminative mindset. But today if you see deep learning is much beyond these uh tasks. Today our focus is just not give for a given input uh classify whether to which class it belongs to. If a patient features are there uh saying whether there is a diss or there is no disses. If there is a time sequence time series sequence is present we want to predict the future value. No, our uh uh aim is just not a prediction or classification. It is just not what is next or which class it belongs to. Then

[11:00] where exactly deep learning today is? So deep modern deep learning is just not about classification or prediction but uh it is increasingly about increasing uh learning the uh data distribution. The task is just not discriminating. The task is to learn the data distribution. Now the question comes why we have to learn the data distribution so that the model can generate a new realistic and more statistically similar data. For example, if you have a task today, you can recall from all the applications that we are seeing around when there is a text uh prompt. When you give a text prompt, take all the GPT um examples.

[11:47] When you give a text prompt, uh today the deep learning models can able to give us an image based on the text. Then if you give uh just a text prompt, it can also give a new paragraph based on what the prompt that we have given. It's just not only a text when we give a damaged data. It can give a complete data. Say for example, I have uploaded my image where it is most of the part of the face is damaged. So it can able to complete that particular image. Next, if you have a noisy data, uh the model can give a cleaned reconstructed data. If you give an existing samples, it can uh synthesize new samples. So, today the modern deep learning is able to do all of these. So, what is the main aim here?

[12:40] The main aim of these modern deep learning models is just not what is the next value or to which class it belongs but here the main is what new data can be created. Now uh so this is all where uh you know from how the transition has happened from discriminative learning to a generative learning. Now from this slide what are we understanding is it not um all we need is to create a new data. This is what uh we

are trying to understand. If this is the only requirement that you know we want to generate a new data. Uh if this is uh the requirement then uh what are the common uh um you know uh concepts that we have already seen in our conventional deep learning models. So if you see if our aim is just to create a new data

[13:37] what about data augmentation right uh the learners here may think so your requirement is to generate the data or to create a data if that is the case why not use a data augmentation because we have seen from our CNN's that data augmentation can always be used to increase the samples. So when we are increasing the samples is it not that we have a data that generated what what is that your thought process on this? You can give a thought of this and then you can look into this uh slide. Data augmentation is definitely one particular fix but what exactly is happening with the data augmentation. We are not creating any new data here.

[14:19] Whatever is the existing data that existing data we are looking uh into different orientations. So probably we are flipping uh uh you take uh one particular image you flip that image. So flip can be done either through horizontal flip or it can be a vertical flip. Then rotation. What is rotation? You will take one image and then you will you will look into this image from a different uh orientations. So that's a rotation zoom either you will zoom out or zoom in then unwanted you're going to crop it and the noise you are going to because your aim is to have a new data so for the existing data can we create some noise so that it is no more that original data can we tell that it's now a new data so what is the

[15:08] thought process so from all of these is it not look into this particular example although I have taken it from an internet source so for a better visualization ation when you see all the images that you are seeing here set of images it is all a same image but looked in a different orientation. So that is what is data augmentation. It is not learning the underlying data distribution here. How the data is from which distribution it has it belongs to.

[15:38] It is not learning anything about the data. It is just visualizing the data uh the image from a different view. Okay. Then you may immediately think what about oversampling. This is also one more technique which is used to increase uh the samples because our aim here is to get a new data. So if oversampling is one of your option then what is the drawback of oversampling? Let's look that with an example. If you see here these are the classes which are um you know where the samples are less in number. So what we are going to do wherever we have an minority uh class samples we are going to increase those samples. So how are we increasing one direct is just duplicating uh um so it may either duplicate or we are going to

[16:29] interpolate. So by doing this we are just improving the count of minority uh class but you are not assuring that there are uh some uh new variant of the existing data that assurance is not been given. So if at all if our thought process is going

in this way that the data augmentation over sampling is anyhow increasing the data then why need why why do we need an uh uh you know generative models. If the thought process is in that direction then the answer is no doubt data augmentation and oversampling are increasing the samples but they are increasing from the existing data only either by different orientation or by duplication. So with this how can we say that it's a new data. Uh so now uh when you take an data

[17:25] and if you duplicate it it will be a similar values. So there is no new learning to the model here. So that is the that is where we are boiling down to an generative model. So generative model models or modeling is mainly aiming to learn the distribution of the data structure of the data so that the new realistic and diverse samples can be generated. It is just not duplication just not variation but we are creating altogether a new data and uh just we are creating a new data that doesn't mean that we are deviating from the original one. No uh even though we are creating a new realistic but still it will be closer to our training data. So uh I hope this is where we have understood that uh data augmentation and

[18:22] oversampling does not help. Now with this let us formally understand what is the definition of a generative modeling. Then you can able to see the difference between discriminative model and generative model. So here the generative model mainly aim to learn the probability distribution of the data. Why it has to learn the distribution of the data? So that its main aim is to create a new samples which are as similar as a training data. Now you can see uh with the definition if you if it is not clear mathematically you can see.

[19:00] So what is uh the discriminative model? Discriminative model is what is the probability that for a given input x it belongs to this particular class. This is what is the prob discriminative models say. Uh but what about a generative model? The generative model is learning this uh distribution. What is the probability of the given uh training samples X [snorts] or it is if at all if you have you know unsupervised data where you don't have labels at all then it is learning the distribution of only the data. Now you have an uh labels also then it is learning P of X uh Y it is learning both input as well as its labels. So let us understand more what exactly is the meaning of these uh two uh probability that we have uh written.

[19:58] Let us understand this in detail now. So the first one P of X what is P of X here? It is an unsupervised generation. What is unsupervised? So here we just have only the data. Look at this. Here is one example where we just have only the data. We have an age. Then we have an height and then width. [snorts] So uh we want to understand for a given height whether for a given age whether height and width is uh I mean height and weight I'm sorry not width height and weight whether it is proper whether it's a normal or abnormal or underweight. So this is what we want to understand. So we don't have any of the uh labels associated with this. All what we have is just the features that is just the data where we have age, height and

[20:49] weight. So here all of this data u we are representing it as x_1 till x_n . So this is x_1 and so on up to x_n . Now all this data belongs to some distribution. Do you all agree to this? So what is the meaning of a distribution? Uh this particular data uh has its own mean and then the uh standard deviation. Correct? So uh we this entire original data uh distribution say let us take this is the original data the distribution of this original X . Now uh what is our aim of a model? The model has to somehow learn this distribution.

[21:34] Now how is that the model learns this distribution? Whenever you take any deep learning model or modern deep learning models for every of the connection we will have the weights associated with that. Let us assume those weights are θ or it can be the distribution parameters itself. It can be μ and uh standard deviation. So what is that? with the help of those parameters θ the model is learning this distribution.

[22:05] So that learned distribution we call it as uh p of x . I hope this is clear to you. So the original whatever the original data that we have that data uh distribution we call it as p data of x . Now uh uh this is our original data distribution. What is the aim? aim is to learn this distribution. How is that the model is learning this distribution? With the help of some parameters and those parameters we are representing it as θ . Okay. Now what is our ultimate aim after training? The ultimate aim is whatever the learned distribution should be as close as original distribution.

[22:50] Right? When the learned distribution and the original distribution are similar then what is that we can able to do from this learned distribution can we now generate a new data right can we now sample uh a new data from this particular distribution so this is what P of X what is a P of X where we don't have labels we are only focusing on here the data only data is given without any labels So and what is the aim? The aim is to learn that underlying data distribution. So that underlying distribution of original data we call it as P of data given uh you know of X . And what is the aim of whatever the algorithm that we are taking uh as part of this discriminative I mean generative uh model here. So the

[23:39] aim of that particular generative model is to learn this distribution. How is that you are learning the distribution? We require some certain parameters. those parameters we are representing it as a θ with the help of parameters it is learning the underlying distribution. Now once the training is completed what is our aim? The learned distribution should be as close as original distribution so that uh when it is close to original distribution now whatever the data that you are going to generate that data will be similar to this original data right. So this is all is this P of X and the second one we have an joint uh modeling. What is joint modeling? Here the model is given both data as well as labels together. So here

[24:32] we have uh the original uh input X and we have an associated label also.

Um we have this. Now what is that the model is going to learn and this entire again once again belongs to some distribution here the aim of generative model is to learn this distribution. Okay you know if so and how it is learning it is learning from the parameter θ . So with the help of these parameters the model is learning the distribution of the given data and here the data is having both uh input that is data as well as the labels associated with it.

[25:18] So here x is the data and y is the label. Now once the model has learned the labels also what else we can perform? So we can also perform the classification here. So although we are discussing about the generating uh data but when we have the labels also with us we have learned the model has learned both data as well as associated label which means now I know if I have a if I have the features like this I can able to classify it to a particular class. So which means it's a classification problem. So in this if at all if the model has to perform the classification it is going to use the base uh theorem. What is uh so is it not that this is discriminative?

[26:04] We have seen from our previous. So what is the probability that the given uh input belongs to this particular uh class y . So is it not that this is a classification problem. So how is that we are able to do a classification problem? We are going to do with the help of base uh theorem. So what what is that important point you need to understand here is although we are using u is generative but we can also do a classification from this and next is a conditional generation what is conditional generation based on a particular condition I don't want all the images to be generated I want only a particular class image to be generated yes even that is also possible so it is what what is the uh possibility that if

[26:53] I have an input X um you know it belongs to this particular uh uh uh label Y [snorts] or a class Y . So it generates the data based on the condition. Cond condition here is you are specifying a particular just generate only a digit seven or generate a smiling face or if you're dealing with an images what if you have less number of malignancy images your your aim is to generate more malignancy images. So in a medical application also you will have an good uh um uh application of this particular concept generate uh conditional generation.

[27:33] Okay. So uh we have understood so far uh that uh we have uh you know either you can have if it is an unsupervised uh if you have a labeled or based on a particular condition you can able to generate it. So now after you get the output. So now say for example you have a decoder or you have a generator. Generator has produced and has given you the output. So that output can be um any of uh this. So what is the first possible output? You can generate an continuous uh real valued values. What does it mean? Suppose if you have a data set where the input is all a real uh continuous real values. Say for example uh you have uh age, height and weight.

[28:25] So here you have all in all these three places you have a numerical value

and you have less samples. Now you want more such a data to be generated. Then what is that? um the model will assume the model is going to give us the uh real valued output but here uh the model is making use of gaussian distribution. So uh here in this particular slide uh try to understand only this concept. If at all if your aim is to generate a continuous real valued uh values then the model is using gaussian distribution. Why gaussian distribution?

[29:03] If you can imagine in the going slides you will be seeing gaussian distribution is a continuous graph. So for continuous data definitely gaussian distribution is a good uh application or a good um u concept to be used. Suppose if our output E is a binary uh say for example uh you are generating you your aim is to generate an binary image or a black and white image where the pixel values are just zero and one the uh you your input image is a black and white image and um and ultimately your output also should be a black and white. So here every pixel that you take belongs to either zero or one class correct. So now whatever the generative models that we are using that generative model assumes a Bernoli distribution. So just only

[29:55] remember try to correlate uh if you have an continuous real value it is gaussian distribution. If if your output is a binary then it is a Bernoli distribution. Just try to remember only these words. You'll be able to connect these in the upcoming slides. Next, what if your output is um you know not a continuous value, not a binary but it has to predict the next word. Uh so you you can take any example. So deep learning is so now the next word the model should predict. So in such a case you will be going for categorical distribution or multinomial distribution. So try to understand that our ultimate aim is to generate the output right uh it has to generate after learning an underlying distribution of

[30:45] data the model has to generate the output. So what are the possible outputs? The possible outputs can be either a real value or it can be an binary or it can be a word. It can be a text application, it can be an image application or it can be an um you know continuous data. So if it is a continuous data gaussian distribution binary bernoli if it is a word it is a categorical distribution. Now uh uh you can uh think um if that is the case in the uh program when we are writing an uh code when we are implementing do we use these words like gaussian barnoli or categorical are we using these words when we are implementing the answer is no. So during implementation we do not usually I mean we do not explicitly

[31:35] write gaussian barnoli or categorical we don't write any of these how is that uh uh the distribution is actually implied here is with the help of activation function and the loss function. So what the different activation functions we use see suppose if the model is assuming a gaussian which means you are producing an continuous values if this is your requirement how is that we write I'm I'm just showing only the few lines of code however you'll be seeing this in the hands-on implementation so how is that we will be uh coding this here the activation what we use is linear activation

definitely output function It should be linear activation right then what should be the loss function the loss function is mean square error MSE

[32:24] so each output value is treated as continuous value then the model predicts the mean of the gaussian distribution right so what although you may not be able to understand at the first session but going forward you'll be able to understand each of these next uh if it is a distribution is a bernoulli then how is that you're going to code it. Suppose if your output is binary then you are going to write activation as a sigmoid and the loss function that we use is binary cross entropy and here each output value is treated as probability of one. Right? What is the probability that the given pixel belongs to zero class or the given pixel belongs to one class. So here the output is treated as the probability and what is the

[33:17] combination sigmoid plus binary cross entropy. Now what is categorical? Categorical is multinomial. You have to predict the next word. So definitely activation is a soft max and um the loss function that we use is categorical cross uh entropy loss. So now we have what is that we have understood from these uh two slides. What we have understood is based on what kind of data that we are going to produce as an output or generate as an output uh what kind of distribution the model is assuming. This is what we have understood. Now where exactly that distribution is coming into the picture.

[34:03] Let us look into this. When we take types of generative models, we have first seen gaussian in the last two slides, right? What is gaussian distribution? The bell uh curve uh this is what is the bell curve for a gaussian distribution. In this what are the two parameters that we will have? We will have the mu and then the standard deviation. These are the two parameters that we have with respect to gaussian distribution. Now uh uh we know that gaussian distribution means you are going to have an continuous values as an uh output. Now look into uh this we have uh one of the most powerful algorithm or model called variational autoencoders.

[34:46] In variational autoencoder where exactly gaussian concept comes into play for now just understand only so much this is a standard uh deviation. Okay. Uh so where exactly you are getting you are taking an input input is passed to an uh encoder and uh encoder main aim is to learn the most important features and then give us the latent space. So uh for this particular first session only try to understand here encoder is there latent space is there in between is where we have the gaussian distribution concept. So in a variational autoencoder uh VAE the encoder maps the input data into the gaussian distribution in the latent space and whatever the new data that we have obtained with the help of decoder. Uh so we are going to

[35:44] generate by sampling from this distribution only uh from this uh distribution whatever is this gaussian distribution. So what you need to understand is now some time back we have understood that to get a real time I mean continuous data model

is assuming the gaussian if model is assuming the gaussian where is that uh assumption coming here so from the encoder to latent space in between is where we have our gaussian distribution concept okay uh uh so uh don't worry if you are not able to relate this because our upcoming weeks Every week we are studying about each model in depth.

[36:26] Second, uh this is a diffusion. Now this is the original input image which is a cat. Now you are adding some noise. Can you able to see some dots from uh how is that? We have added the noise. We have taken the noise from gaussian distribution. We have added some more noise. Some more noise. How long we have added until the image is not at all the back end. The background image here the background the original image is cat but by the time you come here cat is not at all visible the entire image is been you know we have put the entire we have put the noise into this image. So here u um so how this noise has been added is by using a gaussian distribution. So again the word gaussian distribution is coming

[37:11] into the diffusion model and in the reverse process what is happening you are removing the noise and finally you are generating a new sample and the third application if you take so we have an um you know GANs that is generative adversarial networks when you take this GAN what is the role of a GAN uh we have two models generator and discriminator generator has to generate the data discriminator has to discriminate between whether it's a real image or fake you know fake image. If generator has to generate the data, how is that the generator is generating for the first instance? How is that it is generating? So from a um you know from a noise vector which is taken from a gaussian distribution the generator is

[37:59] generating the data. So this slide you need to only connect where all gaussian distribution concept coming into the uh picture. In variational autoencoder we have a gaussian distribution. Uh in diffusion model we have a gaussian distribution where noise is derived or noise is taken from gaussian distribution. In uh GAN generator is generating a data and it is generating from the uh noise from from some vector somewhere something has to start right that something is taken from that gaussian distribution. So this is what you need to understand uh and next we have seen a Bernoli distribution. What is a Bernoli distribution? We have seen that it is mainly used uh if at all if we are generating an um if our aim is an

[38:50] binary uh output. So wherever we have an a binary uh uh data or binary classification application the model is going to consider an barnoli distribution and how barnoli distribution we are going to write at the output layer the activation function and the loss functions are going to differ and next we have an categorical distribution. What is categorical distribution here? Here aim is to predict the next word. So uh in a text generation what is our next word? So uh there we are going to have an categorical distribution and uh we are going to use an um categorical cross entropy and then we will be using uh the softmax activation function. With this we have come

to an end of this particular session. Before we wind up,

[39:41] let us quickly see although we have discussed this all through. Let us quickly see what are the different applications of generative uh uh AI. [snorts] So uh aim is to generate the data right now what all possible data that it can generate. It can either generate the text or it can generate the images or it can generate the audio or the video files or it can generate some code. So if you look into if our aim is to generate the text then we have also seen throughout our particular 12week course what all algorithms that we are seeing we are seeing autoencoders variational autoencoders we are seeing a GAN then we will see uh diffusion models then we will see sequence-to-sequence LLMs then we have uh the

[40:35] multimodal um then in between we have rag So when we have all of these models that we are going to learn if at all if you are dealing with the text applications then among all of these models it is LLM and transformers which are extensively yes transformer you will be studying as part of LLM uh week. So you'll be using these two models extensively when you are dealing with the text application. Suppose if your aim is you're dealing with the images uh um there is a corrupted image you want to generate a new image or a complete image you have less samples you want to generate more samples. So whenever you are dealing with just an images then among these the possible uh application models that we

[41:25] can go is we can use either GAN variational autoencoders or the diffusion models. So using these uh models we can create uh visual applications. Suppose if our aim is speech, music or a sound generation. What exactly are these? Is it not an time sequence here there is an dependency? Whenever we have this kind of an application where you want to generate a new data once again it is the transformers or we can go for GANs or variational autoencoders. And the last if it is a text uh code generation we can use any of these models. So from uh this particular session uh uh we have only touched upon what are the fundamental concepts. We have seen a slow transition from discriminative to generative. Then we have seen what all

[42:22] different probabilistic views we have. Then what all different applications that we can uh generate. So with this we have come to this particular session. Thank you.